2002 tut 6

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Method of Undetermined Coefficients

Particular solution of $ay'' + by' + cy = g(t)$

$g(t)$	$Y(t)$	s
(1) $P_n(t)$	$t^s Q_n(t)$	multiplicity of 0
(2) $P_n(t)e^{\alpha t}$	$t^s Q_n(t)e^{\alpha t}$	multiplicity of α
(3) $P_n(t)e^{\alpha t}\cos\beta t$ or $P_n(t)e^{\alpha t}\sin\beta t$	$t^s\left(Q_n(t)\cos\beta t + R_n(t)\sin\beta t\right)e^{\alpha t}$	multiplicity of $\alpha + i\beta$

Example: Method of Undetermined Coefficients

Example

$$y'' + y' - 2y = t + 1$$

Solution: Characteristic equation: $r^2 + r - 2 = 0$, roots $r_1 = -2$, $r_2 = 1$.

$$y_h = c_1 e^{-2t} + c_2 e^{t}$$

Corresponding to case (1) $s = 0$
Assume $Y(t) = At + B$:

$$Y'' + Y' - 2Y = t + 1 \Rightarrow A = -\frac{1}{2}, \quad B = -\frac{3}{4}$$

Final solution:

$$y = c_1 e^{-2t} + c_2 e^{t} - \frac{1}{2}t - \frac{3}{4}$$

Exercise

Exercise

$$y'' + y' - 2y = (t+2)e^{-2t}$$

$$y'' + y' - 2y = -10sint$$

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$$y'' + y' - 2y = (t+2)e^{-2t}$$

$$y'' + y' - 2y = -10sint$$

Solution

The first one corresponding to case (2), $\alpha = -2, s = 1$
The answer is $Y(t) = (-\frac{1}{6}t^2 - \frac{7}{9}t)e^{-2t}$
The second one corresponding to case (3), $\alpha + i\beta = 0 + i, s = 0$
The answer is $Y(t) = cost + 3sint$

Variation of Parameters

Consider

$$ay'' + by' + cy = g(t)$$

The method of undetermined coefficent only works for special $g(t)$,but by the method of variation of parameter, we can pove that

Theorem

Given $y_1(t)$ and $y_2(t)$ as solutions to the homogeneous equation, a particular solution to the nonhomogeneous equation is:

$$Y(t) = -y_1(t) \int \frac{y_2(t)g(t)}{W(y_1, y_2)(t)} dt + y_2(t) \int \frac{y_1(t)g(t)}{W(y_1, y_2)(t)} dt$$

where $W(y_1, y_2)(t)$ is the Wronskian.

Example: Variation of Parameters

Example

$$y'' + 4y = \frac{3}{\sin t}$$

Solution: Homogeneous solutions: $y_1 = \cos 2t$, $y_2 = \sin 2t$.

$$W(y_1, y_2)(t) = 2$$

$$\int \frac{y_2(t)g(t)}{W(y_1, y_2)(t)} = \frac{6cos(t)}{2} = -3sin(t)$$

$$\int \frac{y_1(t)g(t)}{W(y_1, y_2)(t)} = \frac{3}{2} \ln |\tan \frac{t}{2}| + 3\cos t$$

Particular solution:

$$Y(t) = -3 \sin t \cos 2t + \left(\frac{3}{2} \ln |\tan \frac{t}{2}| + 3 \cos t \right) \sin 2t$$

Excersice

Exercise

Solve

$$y'' - 3y' + 2y = \frac{e^{3t}}{e^t + 1}$$

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Solve

$$y'' - 3y' + 2y = \frac{e^{3t}}{e^t + 1}$$

Solution

$y_1 = e^t, y_2 = e^{2t}$
$W(y_1, y_2)(t) = e^{3t}$
$u_1 = \int -\frac{gy_2}{W} = -\int \frac{e^{2t}}{e^t+1} = -e^t + ln(e^t + 1)$
$u_2 = \int \frac{gy_1}{W} = \int \frac{e^t}{e^t+1} = ln(e^t + 1)$

The general solution is:

$$y = c_1y_1 + c_2y_2 + Y(t) = c_1e^t + c_2e^{2t} + e^t(-e^t + ln(e^t + 1)) + e^{2t}ln(e^t + 1)$$

Reduce the order

Reduction of order: Suppose $y_1(t) \neq 0$ is a solution of $y'' + p(t)y' + q(t)y = 0$, find y_2.
Method I. Abel's theorem.

$$w(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = ce^{-\int p(t)dt} = e^{-\int p(t)dt}$$

$\Rightarrow \quad y_1 y_2' - y_1' y_2 = e^{-\int P(t)dt}$, first order linear ODE for y_2.
Method 2. Variation of parameter. Let $y_2(t) = v(t)y_1(t)$.

$$\Rightarrow y_2'' + py_2' + qy_2 = v''y_1 + v'\left(2y_1' + py_1\right) = 0$$

Although it's a second order linear ODE for v, it can be regarded as a first order linear/separable ODE for V'.

Example

Example

Consider $2t^2y'' + 3ty' - y = 0, t > 0$. $y_1 = t^{\frac{1}{2}}$ is given, find y_2 s.t. y_1, y_2 form the fundamental set of solutions.

Method 1: See the last tutorial note.

Method 2: Let $y_2(t) = v(t)t^{\frac{1}{2}}$,

$$
\begin{aligned}
& y_2'' + \frac{3}{2t}y_2' - \frac{1}{2t^2}y_2' = 0 \\
\Rightarrow \quad & v''t^{\frac{1}{2}} + v'\left(2 \cdot \frac{1}{2}t^{-\frac{1}{2}} + \frac{3}{2t}t^{\frac{1}{2}}\right) = 0 \\
& v'' + \frac{5}{2t}v' = 0 \\
& v' = c_1 t^{-\frac{5}{2}} \\
\Rightarrow \quad & v = -\frac{2}{3}c_1 t^{-\frac{3}{2}} + c_2
\end{aligned}
$$

Take $v = t^{-\frac{3}{2}}, \quad y_2 = vy_1 = t^{-1}$

Exercise

Exercise

$t^2y'' - t(t+2)y' + (t+2)y = 0, \quad t > 0$

$y_1 = t$, *find* y_2.

Exercise

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$t^2y'' - t(t+2)y' + (t+2)y = 0, \quad t > 0$

$y_1 = t$, *find* y_2.
Sol:$y_2 = te^t$